Screen Use and Mental Health in Adolescents and Emerging Adults

A White Paper :

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Section 1: Framing the Question

The question: To what extent do the amount and the manner of screen use each affect the mental health of individuals aged 13 to 24?

Most public arguments about screens are conducted in a single unit: hours per day. That unit is close to useless on its own. It counts a video call with a grandparent and an hour of algorithmically served short-form video as identical inputs, and it treats a teenager who watches two hours of television as equivalent to one who checks a feed two hundred times across the same span. Our question is built to test whether that unit is measuring the right thing.

We are not the first to doubt it. In January 2026 the American Academy of Pediatrics released its first major revision to screen-time guidance in a decade and abandoned hour-based limits almost entirely. The policy statement tells practitioners that if families want a number, one can be discussed to fit their routine, but that the most important considerations are content quality and prioritizing sleep, play, physical activity, and reading.[1] The leading pediatric body in the country has concluded that duration is the wrong primary measure. This paper examines whether the evidence supports that conclusion and what follows from it.

Hypothesis

We expect that the manner of screen use predicts mental health outcomes substantially better than the amount of screen use, and that the weakness and inconsistency of existing findings is largely a product of measuring duration instead of pattern.

This is a falsifiable claim and we intend to treat it as one. If the data show that total duration predicts depressive symptoms, anxiety, and sleep disruption as well as or better than compulsive-use measures, our hypothesis is wrong and we will say so. Section 5 is deliberately structured around what would follow if the hypothesis holds; Section 4 is structured around what would show that it does not.

Definitions

Amount is the total duration of screen use, whether self-reported, parent-reported, or captured passively by a device. Nearly all existing data and nearly all existing policies use this measure.

Manner is the pattern of use rather than its quantity. We operationalize it using the nine-component framework behind the Social Media Disorder Scale, which the international HBSC study (detailed in Section 4) uses to classify adolescents: preoccupation, withdrawal, tolerance, unsuccessful attempts to cut back, displacement of other activities, continued use despite problems, deception about use, use to escape negative mood, and loss of a relationship or opportunity. The scale has been validated across 222,532 adolescents in 44 countries, which means the construct is not an artifact of a single study or population.[2]

Mental health means depressive and anxiety symptoms, stress, sleep disruption, and self-reported wellbeing.

Scope and justification

We examine individuals aged 13 to 24. This is not a single homogeneous population and we do not analyze it as one. It comprises two cohorts, adolescents roughly 13 to 17 and emerging adults roughly 18 to 24, which differ in developmental stage, legal status, living situation, and platform access. Every finding in Section 4 is reported against the specific dataset and age band that produced it. No measure is averaged across the two.

We include both because the boundary between them is where our hypothesis is testable. There is precedent for treating this span as a developmental unit. Sawyer and colleagues argued in *The Lancet Child & Adolescent Health* that adolescence is better defined as 10 to 24 than 10 to 19, given earlier puberty, continued neurological development into the twenties, and the delayed timing of adult role transitions, and they specifically identify digital media as a force acting across the entire span.[3] That proposal remains contested. Correspondence in the same journal argued it blurs a useful distinction between adolescence and young adulthood.[4] We do not need it settled. We cite it to establish that the range is a recognized object of study rather than an arbitrary bracket.

Our own reason is narrower and specific to the hypothesis. Age 18 is the point at which every externally imposed limit on screen use ends at once: parental controls, school device policies, and household rules all lapse simultaneously. If duration-based interventions produce durable self-regulation, that should be visible on the far side of that boundary. If they only suppress use while enforced, the discontinuity should be visible too. Restricting the paper to under-18 would remove the only natural experiment our evidence base contains.

Positionality

We are all heavy screen users, which is a reason for caution rather than a claim to authority. Being inside the problem can push us to downplay it just as easily as being outside it pushes critics to exaggerate. So we hold our own assumptions to the same standard as the data: no flattening everything into hours per day, a clear line between compulsive and intentional use, and no treating correlation as causation because the causal version is a better story.

We will credit every source and link the raw data. We will interview experts and ordinary heavy users, whose perspective is usually missing from clinical research, and we will look for voices that complicate

the "screens are destroying a generation" framing rather than only those that confirm it. CU faculty, including researchers at the CU Change Lab, will be consulted throughout.

The risk of getting this wrong is not just a bad grade. Overstated findings get used to justify regulation built on weak evidence; sloppy findings get used to dismiss real mental health concerns. Being careful is what keeps this from being turned against the people it is meant to help.

Limits on the evidence

First, our sources measure screen usage in four different ways, and none measures it directly. The method used by each dataset is detailed in Section 4.

Second, the measured effect of screen use on wellbeing is small enough that serious researchers argue it is not worth acting on. A specification curve analysis across three large datasets found the association between digital technology use and adolescent wellbeing was negative but small, explaining at most 0.4% of the variation, and concluded the effects were too small to warrant policy change.[5] This is the strongest available objection to our premise. We take it seriously, and Section 4 explains why we think it is also evidence for our hypothesis rather than against it.

Third, none of our datasets can establish causation. All are observational and self-reported. None controls confounding variables in the way an experimental design would, so where this paper describes a mechanism it is drawing on longitudinal work by others and says so.

Section 2: History of the Issue

Television

Television was the first medium that could reach children without passing through a parent or a teacher. The worry arrived almost immediately. By 1950 a network executive was publicly assuring parents that television was not an "electronic baby sitter," a phrase he would not have needed to deny unless it was already in wide circulation.[1] That matters for mental health long before anyone measured it, because the earliest public concern about screens was never about hours. It was about what unsupervised exposure was doing to children's development, and about adults losing the ability to see it happening.

Proof that young audiences can be captured

Cigarette brands sponsored primetime programming children watched, including *The Flintstones*, until television advertising for cigarettes was banned in 1971. The effort then moved to print, billboards, and merchandising for another two decades.

The proof that it worked came in 1991. Researchers asked 229 preschoolers aged three to six to match logos to products and found Old Joe the Camel was the most recognized logo in its category. Ninety-one percent of six-year-olds identified it correctly, nearly matching Mickey Mouse.[2] Camel's share among underage smokers rose from 0.5% to 32.8% over the campaign's run. Throughout, the industry denied targeting children.

The relevance here is mechanistic, not moral. A cartoon built familiarity and trust with children years before any purchase decision. That is the same pathway Section 3.4 describes running through influencer sponsorship today, and it is the pathway most often proposed to connect screen use to anxiety and depressive symptoms.

The regulatory response and its limits

Organized opposition formed early. Action for Children's Television, founded in 1968, petitioned the FCC in 1970 to ban advertising from children's programming and in 1977 petitioned the FTC to prohibit advertising aimed at children too young to understand selling intent.[3] In 1978 the FTC took up the question in a rulemaking known as "kidvid," proposing to ban advertising directed at children too young to grasp its purpose and to ban high-sugar-product advertising to children eight to eleven.[4]

The industry response was decisive. Congress allowed the agency's funding to lapse, the FTC shut down briefly, and 1980 legislation stripped its authority to make broad rules on the basis of unfairness.[5] The proceeding closed in 1981 without a rule.

Regulation eventually came in narrower form. The Children's Television Act of 1990 capped commercial time at 10.5 minutes per hour on weekends and 12 on weekdays, required educational programming, and restricted host-selling.[6] Reach in the affected categories fell afterward, confirming the practices had worked as critics claimed.

Two lessons carry forward. Regulating children's media is politically dangerous for the regulator, which the FTC has not forgotten. And when regulation did arrive, it counted minutes. That is a duration limit, the same unit every intervention has used since, and the unit this paper questions.

The digital era

The 1990 rules do not transfer. Age gates are easily defeated, platforms disclaim responsibility for what circulates, and behavioral data is used to target content back at the same users. The structural difference is that advertising is no longer a separable block of time that can be capped. It is woven into content engineered to hold attention.

Recent cases show the pattern. The prediction market Polymarket ran undisclosed creator partnerships featuring simulated trades across more than 1,100 videos with roughly 140 million views.[7] A watchdog complaint alleged a MrBeast sweepstakes collected names, addresses, and phone numbers from participants including children under 13.[8] Both work by converting a trusted relationship into sustained engagement, which is the same structure the Surgeon General's 2023 advisory identifies when it names endless scrolling, notifications, and engagement-optimized algorithms as drivers of the excessive use associated with sleep disruption, social comparison, and body image concerns.[9]

What the history establishes

Seventy-five years of concern about children and screens has produced one durable finding and one durable failure. The finding is that media built to hold young attention does hold it, reliably, in ways children cannot detect. The failure is that every response has been written in units of time, minutes per hour in 1990 and hours per day today, while the thing being regulated shifted from advertising blocks to the design of the content itself.

Whether that produces measurable mental health harm is the question this paper tests rather than assumes. What the history shows is that the tools built to address it have consistently measured the wrong thing.

Section 3: Stakeholders

3.1 Adolescents and emerging adults, ages 13 to 24

Position. This group is the one the question is about. ATUS data shows 15-to-17-year-olds average close to three hours of screen time daily (Section 4.1). Adolescence is a sensitive developmental period and the point at which many mental health conditions first appear, which means changes in this population register more sharply than they would in adults.

Interests. Connection, creativity, and information access are real benefits, not rationalizations. Pew found 74% of teens say social media makes them feel more connected to friends and 63% say it gives them a place to be creative. Any intervention that removes these without replacing them imposes a cost this group will resist.

Incentives. Mixed and internally contradictory. Pew reports 45% of teens say they spend too much time on social media, up from 36% in 2022, and 48% think social media is mostly negative for people their age, but only 14% think it is mostly negative for themselves.[1] That 34-point gap is one of the more informative numbers in this paper. It is roughly what you would see if harm were concentrated in a minority rather than distributed evenly, and it is also what you would see if people systematically exempt themselves from problems they recognize in others.

Limitations. This group has essentially no leverage over platform design and cannot meaningfully opt out, since the platforms are where their social lives are. Individual abstention carries a real social cost, which makes it an unstable solution. They are also the stakeholder least able to assess their own use accurately, which is not a criticism of them specifically, since self-report diverges from logged use across every population studied.

3.2 Parents and teachers

Position. Historically the first to notice when something is wrong with a young person, and uniquely able to observe pattern rather than duration. A survey records that a teenager spends four hours a day on a phone. A parent sees whether that teenager can put it down.

Interests. Teachers want students who can sustain attention long enough to learn, and educators report students struggling to retain material and reason independently as short-form content has spread.[2] Parents want their children prepared for adulthood. Roughly 60% of public-school parents report worrying about technology's effect on their child's wellbeing.[3]

Incentives. Both groups are held accountable for outcomes they do not control. Teachers are evaluated on student performance; parents are judged on their children's development. That creates pressure to act even where evidence about what works is thin, which is worth noting because it makes both groups receptive to interventions that feel decisive rather than interventions that are effective.

Limitations. This is the crucial point and the one most often missed. Parents and teachers have the best view of the problem and the least capacity to address it. A household rule affects one household. A

classroom policy affects one classroom. Some states have moved on school device restrictions, such as Iowa House File 2676,[4] but coverage is partial and the statutes are loosely drafted. Neither group can change how a product is designed, and the AAP's 2026 revision implicitly acknowledged that the standards previously handed to parents had become impossible to meet.[5]

3.3 Social media companies

Position. Meta, TikTok, and Snap make the decisions that define the environment: recommendation algorithms, notification schedules, advertising load, age restrictions, and the features determining session length. Up to 95% of young people aged 13 to 17 use a social media platform, nearly two thirds daily, and one third almost constantly.[6] [7]

Interests. Revenue. Meta reported roughly $200 billion for 2025, with a substantial share of users in younger brackets.[8]

Incentives. More time on the platform means more impressions, more consumption, and more interaction. Critically, the model does not optimize for total hours so much as for return frequency and session depth. That is to say it optimizes for precisely the pattern of use our hypothesis identifies as harmful, and is largely indifferent to the duration metric every intervention measures.

Limitations and constraints. Companies do face pressure, and have responded. Meta's teen accounts apply protections automatically and require parental permission for under-16 users to loosen them.[9] Snapchat uses private-by-default teen accounts. In August 2026 Meta settled with a coalition of states for a reported $17 billion over allegations of addictive design and inadequate child safety protections.[10] But the binding constraint on these companies is fiduciary, not reputational. A settlement is an expense. Until a safety measure costs less than the liability it prevents, the rational move is to pay.

3.4 Advertisers and brand sponsors

Position. Toy companies, food brands, and edtech apps buying presence in children's media, and more consequentially the brands paying children's creators directly through sponsorship rather than platform-served ads. The American Academy of Pediatrics puts annual spending on advertising to children at roughly $250 billion.[11] For many children's channels, sponsorship is now the largest single revenue source, exceeding platform payouts.[12]

Interests. Low-cost access to an audience that trusts the messenger more than it questions the message.

Incentives. This is the mechanism that matters. Sponsorships are priced on watch time, retention, and engaged views rather than on impressions.[13] Once a creator's income depends on how long a young viewer stays, the content adapts: longer videos, multi-part formats, cliffhangers, series structures. No individual creator needs to intend manipulation for this to happen, because the payment structure rewards whoever holds attention longest. What is being purchased is not ad space but an existing parasocial bond, the one-sided relationship a viewer forms with a creator they watch repeatedly, which functions much like a friendship. Children with a strong parasocial bond showed little change in attitude toward a sponsored

product even after understanding the video existed to sell something,[14] and a majority of adolescent subscribers report relating to influencers more than to traditional celebrities.[15]

Evidence of scale. A 2021 study in *Pediatrics* found many popular kid-influencer videos contained undisclosed placements for brands including McDonald's, Hershey's, and Taco Bell,[16] and agencies such as Pocket.watch's Clock Work division exist specifically to run this model across a roster of child creators.[17] This was once a small business, with one family creator describing a first brand deal paying $500,[18] before it scaled into a multibillion-dollar industry with its own agencies.

Limitations and accountability. The industry polices itself through the Children's Advertising Review Unit, which did not formally extend its guidelines to influencer marketing until 2020 and is funded and run by the industry it oversees.[19] Advertisers are in conflict with pediatric researchers, parent advocacy groups, and the FTC, all of whom argue that a watch-time payment model creates structural pressure toward more screen use that disclosure law has not caught up with. Their relationship with creator families is more complicated: a genuine financial partnership, but one critics argue pressures families to keep content engaging enough to sustain the income.

3.5 Public health institutions and regulators

Position. The AAP, the APA, the Surgeon General's office, the FTC, and Congress set the standards everything else is measured against.

Interests. Defensible guidance that can be followed and enforced.

Incentives. Institutions need recommendations simple enough to communicate. "No more than two hours" is communicable; a nine-item behavioral construct is not. This is a significant part of why duration became the standard unit, and it is a reason to expect institutional resistance to any measure that is more accurate but harder to explain.

Limitations. The FTC has been cautious about children's media regulation since 1981 for reasons Section 2 describes. Congress has passed nothing: KOSA cleared the Senate 91 to 3 in 2024 and has not moved in the House. Institutions can revise guidance faster than they can change law, which is exactly what the AAP did in 2026, and which leaves a widening gap between what clinicians now recommend and what any statute requires.

Section 4: Data and Evidence

4.1 American Time Use Survey

The ATUS was recommended to us by Jennifer Schwartz. It is sponsored by the Bureau of Labor Statistics and conducted by the Census Bureau, and records how Americans spend their time.[1] Interviewers ask participants to recall the previous 24 hours, including duration, location, and how they felt about each activity. This gives us both when respondents are online and some information about the form that screen time takes. The dataset is very large, and screen-related activity is a narrow subset we extract.

We are working with two files. The **Activity File** is in long format, one record per activity per respondent, useful for timing, duration, and sequence. The **Activity Summary File** contains one row per respondent with total duration per activity type, useful at a higher level.

Variable	Official classification	What it captures
t120303	Television and movies (not religious)	Television, Netflix, Hulu, YouTube, streaming films
t120304	Television (religious)	Religious television programming
t120307	Playing games	Video, computer, and board games; cards; puzzles
t120308	Computer for leisure (excluding games)	Recreational browsing, social media, personal-interest computing

Preliminary findings for ages 15 to 17: an average of 2 hours 57 minutes of screen time daily, comprising 81 minutes of television and movies, 72 minutes of games, and 24 minutes of leisure computer use. Total screen-related time rises to 3 hours 43 minutes on weekends and holidays.

Measurement method. Twenty-four-hour recall diary, self-reported to an interviewer.

Limitation relevant to our hypothesis. ATUS measures duration and activity category, and folds social media into general leisure computer use. It describes the landscape. It cannot test amount against manner.

4.2 National Survey of Children's Health

The NSCH is collected annually by the Census Bureau on behalf of HRSA's Maternal and Child Health Bureau, a federal partnership rather than a private or advocacy-funded effort.[2] It tracks the physical and emotional health of children aged 0 to 17 nationally and by state. Since 2016 it has used mailed and web-based questionnaires sent to randomly selected households, with samples in the tens of thousands annually, roughly 30,000 in 2018 and 42,000 in 2020. Raw files are published openly with codebooks and questionnaires.

Because it is not advocacy-funded, the NSCH carries strong credibility as a neutral source. It is not designed to prove a conclusion about screen time. Government involvement does mean the questions reflect institutional priorities, and AAP screen-time guidelines are the clearest expression of those priorities. This matters more than it first appears, because the AAP revised its assumptions in 2026.[3] NSCH items written against the old two-hour standard now encode a benchmark their own source has retired.

Researchers have used NSCH data to link screen time to obesity, psychological wellbeing, and family media rules. One study found 49% of children exceeded two hours daily, with lower family-meal frequency and absence of television rules associated with higher screen time.

Measurement method. Parent or caregiver proxy report.

Limitation. Entirely parent-reported and does not distinguish content format, so it addresses total time only.

4.3 APA Stress in America

Run annually by the American Psychological Association since 2007, sampling adults plus a smaller teen panel. Coverage of the 2018 wave, which appears representative of the survey's structure, reported roughly 3,500 adult interviews and 300 with teenagers aged 15 to 17.[4]

This is our primary source for self-reported mental health and for attitudes toward social media over time. Its distinctive value is the generational comparison, since it can show whether young people differ from older cohorts rather than merely that they report distress. The same reporting found Gen Z describing the worst mental health of any generation surveyed, with just 45% rating their mental health excellent or very good against 70% of Boomers. On social media specifically, roughly half called it a source of support while 45% called it a stressor. That split, within one population, is the pattern our hypothesis predicts.

Measurement method. Respondent self-report.

Limitations. The teen sample is small relative to the adult sample. It measures self-reported stress, not clinical diagnosis. And its screen items are attitudinal rather than behavioral, so it captures how young people feel about social media rather than how they use it.

4.4 ACHA National College Health Assessment

A national survey of college students run by the American College Health Association, with roughly 300 questions covering health status, risk and protective behaviors, access to health information, impediments to academic performance, and mental health.

This covers the emerging-adult end of our range and includes a direct screen-use item. The Fall 2024 administration found nearly 60% of students reported six or more hours of social media in a typical week, more than 75% reported fewer than eight hours of sleep on weeknights, and 30% reported anxiety negatively affecting their academics.[5]

Including college students matters because we should not assume effects are uniform across a twelve-year age span. Parental controls, school policies, and household rules all vanish at once at college, which makes this population a natural test of whether limits imposed in childhood produce durable self-regulation or merely suppress use while enforced.

Measurement method. Respondent self-report.

Limitations. Access is arranged through ACHA directly rather than open download.[6] The sample is college students specifically, excluding non-enrolled 18-to-24-year-olds. And the survey is administered by participating institutions rather than drawn as a national probability sample, so we will identify which universities are represented and flag likely bias.

4.5 AAP screen-time guidance

The AAP is not a dataset. We include it because it functions as the standard American screen-time policy, survey design, and parental practice have all been calibrated against, and because it changed.

The 2016 guidance recommended no screen media under 18 months, one hour daily of high-quality programming for ages two to five, and consistent limits thereafter.[7] The 2026 revision abandons the hour-based frame in favor of content quality and displacement, with any numeric limit treated as a conversation to be fitted to a family rather than a standard.[3]

Our question asks whether amount or manner is the operative variable. In January 2026 the leading pediatric authority in the United States answered in favor of manner and rewrote its guidance accordingly. This is not something we can test our hypothesis against. It is evidence that the hypothesis is becoming clinical consensus, and it defines the gap this project can fill, since every dataset above was designed against the prior standard.

4.6 Supporting evidence from outside our datasets

Three findings from the longitudinal literature inform our interpretation, and we flag them as external rather than as data we hold.

A UNC study followed 169 North Carolina middle schoolers for three years and linked habitual checking to increased neural sensitivity when anticipating social rewards and punishments, a pattern consistent with compulsive behavior rather than with volume of use.[8]

A JAMA study tracked 4,285 children over four years and found that baseline screen time did not predict later negative mental health outcomes, but that addictive use did: feeling unable to stop, distress when separated from a device, and using it to escape problems. Lower-income participants and some racial groups were more likely to fall into those patterns.[9] This is the single most direct test of our hypothesis in the existing literature, and it supports it.

The 2023 Surgeon General's advisory, synthesizing meta-analyses rather than presenting new data, concluded that social media may cause and perpetuate body image concerns and affect sleep quality and

social comparison, while stating that the evidence is not yet sufficient to determine whether social media is safe for children.[10] It specifically names design features, including endless scrolling, notifications, and engagement-optimized algorithms, as drivers of excessive use.

4.7 The disagreement, and why we read it as we do

The strongest objection to this paper's premise is the finding that digital technology use explains at most 0.4% of variation in adolescent wellbeing across 355,358 participants, an effect the authors argued was too small to justify policy change.[11] When critics re-specified the analysis, the authors re-ran it and reported the median association was not significantly different, noting that wearing glasses showed a more negative association with adolescent wellbeing than technology use did.[12]

We do not dispute that finding. We think it is the expected result of measuring the wrong variable. A literature built on duration, where duration is a weak predictor and pattern is a strong one, would produce exactly this signature: real effects, small, unstable across specifications, with sustained disagreement about whether they matter. The JAMA result, null for duration and positive for compulsive use in the same cohort, is what that looks like when both variables are measured at once.

The instrument that would let us test this directly already exists. The Social Media Disorder Scale used by HBSC classifies adolescents by pattern rather than hours and has been validated across 44 countries.[13] No major American dataset has adopted it. That absence is the gap this project is positioned to address.

Section 5: Potential Solutions

5.1 What has been tried

Solutions have come from two places that rarely coordinate: platforms building features voluntarily, and governments attempting to mandate them.

The regulatory floor predates the problem. COPPA passed in 1998, when roughly a quarter of American households had internet access. It restricted data collection from children under 13 and said nothing about design, algorithms, or teenagers. Given the history in Section 2, that is not an accident of timing so much as the high-water mark of what was politically achievable after 1981.

Opt-in awareness tools, 2018. Apple's Screen Time and the Instagram and YouTube usage dashboards were the first modern attempt: opt-in, easily circumvented, aimed at nothing in particular. Parent forums are full of workarounds,[1] which is unsurprising, since a company whose revenue depends on time in the app has no incentive to make its own limits binding. **Verdict: failed.** Their value was normalizing the idea that time on app is worth tracking.

Default-on limits, 2023. TikTok made its 60-minute cap for under-18 users the default rather than a setting.[2] Defaults determine most behavior, so this mattered more than the previous wave, but teens could still switch it off. **Verdict: partial.**

Parent-enforced controls, 2024 to 2025. TikTok's Family Pairing lets a parent set limits the teen cannot override, and Meta's teen accounts require parental permission for under-16 users to loosen protections.[3] **Verdict: mechanically effective, practically limited.** Enforcement moves onto parents who often lack the technical fluency, it introduces an adversarial dynamic into the relationship the intervention depends on, and Section 4.4 raises the question of what happens when these simply expire at 18.

Legislation. KOSA, introduced in 2022 after the Facebook whistleblower disclosures, is the first serious attempt since COPPA to impose a duty of care on platforms, including restrictions on addictive design and algorithmic targeting of minors. It passed the Senate 91 to 3 in 2024 and has not passed the House. A parallel effort to raise COPPA's age threshold to 16 and ban targeted advertising to minors sits in the same place.[4] The Kids Off Social Media Act belongs to the same wave.[5] **Verdict: untested.**

Litigation. Meta's $17 billion settlement in August 2026 is the largest action to date.[6] **Verdict: unclear.** The Joe Camel precedent suggests litigation can end a campaign without ending the practice behind it.

Every implemented solution regulates duration. The only proposals targeting design are the ones that have not passed. If our hypothesis holds, the entire deployed toolkit has been aimed at the variable that predicts less.

The reason is not that anyone chose badly. Duration is easier to measure, easier to legislate, and easier to explain to a parent than a nine-item behavioral construct, a constraint Section 3.5 identifies as structural to how institutions work. But an intervention that removes thirty minutes a day while leaving a compulsive pattern intact has not touched the thing that predicts harm.

5.3 The Haidt proposal, and where its evidence comes from

No discussion of solutions in this space can skip Jonathan Haidt, because his is the proposal currently shaping public debate. Haidt, a social psychologist at NYU Stern, argued in *The Anxious Generation* that a "great rewiring" of childhood beginning around 2012, when smartphone adoption crossed 50%, produced an epidemic of adolescent mental illness.[7] He proposes four norms: no smartphones before high school, no social media before 16, phone-free schools, and substantially more unsupervised play.

Where the data comes from. Haidt does not generate new data. He synthesizes existing datasets, primarily Monitoring the Future, the National Survey on Drug Use and Health, CDC mortality and emergency-department records, the General Social Survey, and internationally the HBSC study, which is the same instrument Section 1 draws its problematic-use construct from. Alongside the book he maintains public collaborative review documents with Zach Rausch and Jean Twenge, which compile the studies and datasets underlying the argument and are open for anyone to inspect.[8]

The transparency is a genuine strength. Publishing your evidence base in an editable public document, before critics ask, is more than most authors in this space do.

The criticism. It is substantial and comes from the researchers whose work Haidt draws on. Candice Odgers, reviewing the book in *Nature*, wrote that hundreds of researchers including herself have looked for effects of the size Haidt describes and found a mix of null, small, and mixed associations, mostly correlational, and that where longitudinal associations appear they often suggest young people already struggling use these platforms more rather than the reverse. She concluded that the book is a story in search of evidence.[9] Andrew Przybylski characterized the method as vote-counting, treating a tally of supporting studies as decisive regardless of individual study quality, and argued that extraordinary claims require extraordinary evidence Haidt has not produced.[10] The 0.4% effect-size finding in Section 4.7 is the quantitative core of that objection.[11]

Where we land. Haidt's diagnosis and his prescriptions come apart, and that separation is this paper's contribution to the debate. The causal claim, that smartphones produced a mental health epidemic, outruns its evidence, and we are not prepared to assert it. But two of his four proposed norms, phone-free schools and delaying social media to 16, are among the only widely discussed interventions that target when and whether rather than how many hours, which makes them better aligned with what the evidence actually supports than most of the toolkit in Section 5.1. It is possible to think Haidt has overstated the case and still think several of his remedies point in the right direction, and we do.

5.4 What would follow if our hypothesis holds

If manner predicts outcomes and amount largely does not, three things change.

Measurement comes first. No intervention can target compulsive use while no major American dataset measures it. The HBSC instrument exists and is validated across 44 countries.[12] Adopting it, or something like it, in NSCH, ATUS, or the ACHA-NCHA is the prerequisite for everything else. This is the least dramatic recommendation in this paper and probably the most important one.

Design regulation beats time regulation. KOSA's duty-of-care framing is the only current proposal built on the right variable. Whatever happens to that specific bill, the design-based approach is the one worth defending.

Screening should replace universal limits. If roughly one in ten young people show problematic-use patterns and most heavy users are fine, then blanket hour caps impose costs on the majority to reach a minority they may not even help. The AAP's 2026 guidance already moves this direction, directing clinicians to screen when use affects sleep, relationships, and functioning.[13] That is a pattern criterion in clinical practice, and it is the model.

5.5 Our deliverables

The white paper is the research stage. The advocacy stage needs a format, and we are weighing two.

Option A: a written piece on Substack or Medium. Fast to produce, easy to distribute, and well suited to the argument, which is fundamentally verbal: the wrong variable is being measured, here is why, here is what to measure instead. Substack is the better of the two, since it builds a subscriber list we would own rather than renting attention from a platform's recommendation algorithm, a distinction Section 3.4 should make us sensitive to. The risk is that a written argument about screen time competes directly with *The Anxious Generation*, a bestseller with a two-year head start.

Option B: an interactive website. More ambitious and better matched to the material. The central claim is a comparison between two variables, and a reader who can move a slider between hours per day and problematic-use score and watch which one moves the outcome has understood the argument in a way no paragraph achieves. The four-category HBSC breakdown of non-active, active, intense, and problematic users is naturally visual, and the fact that most heavy users are fine is far more persuasive shown than stated.

Our recommendation: build both, in that order. The written piece is the argument; the site is the demonstration. Publishing the essay first tests whether the framing lands and produces an audience to send to the site later. Building the site first risks spending the semester on interaction design for an argument we have not yet pressure-tested. The essay also forces the prose to be right, which the site will need anyway.

One constraint worth stating now. The interactive version requires data we do not yet have. The visualization that would make the case depends on a dataset measuring both amount and manner in the same population, and Section 4 establishes that none of ours does. Either we secure ACHA-NCHA access and add our own instrument, or the interactive piece visualizes HBSC's European data with an explicit note about generalizability. That decision should be made before any development work starts.

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Section 1: Framing the Question

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